KIRUTHIKKUMAAR RADHAKRISHNAN

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SUMMARY

Results-driven Artificial Intelligence specialist and full-stack developer with a proven track record in engineering advanced computer vision and deep learning models. Proficient in bridging complex neural networks (CNNs, RNNs, ANNs) with scalable web architectures using Python, ReactJS, and Flask. Adept at delivering high-accuracy recognition systems and optimized backend solutions to drive impactful, real-world tech innovations.

TECHNICAL SKILLS

- Programming Languages: Python, Java, C , JavaScript, HTML, CSS.
- AI & Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN) , Long Short-Term Memory (LSTM), Natural Language Processing (NLP), Natural Language Understanding (NLU).
- Machine Learning & Computer Vision: K-Means Clustering, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), HOG, RetinaFace, Deep Metric Learning, EasyOCR.
- Web Frameworks & Architecture: ReactJS , Flask, RESTful API Development.
- Tools & Environments: VS Code, Jupyter Notebook, Android Studio, Arduino.
- Core Competencies: Full-Stack Web Development , Internet of Things (IoT) , Object & Face Recognition Systems , Predictive Modeling , Problem Solving

EXPERIENCE

Leanfolks Technology Solutions (Erode, India) November 2024 – August 2025

Role: Full Stack Web Developer Intern

- Spearheaded the full-stack development of NOVARACE, an in-house proprietary race management platform, building a responsive ReactJS frontend and robust Flask backend to serve thousands of end-users.
- Engineered a sophisticated Face Recognition System for NOVARACE using HOG, Linear SVM, and Deep Metric Learning (ResNet), achieving 98%+ accuracy in participant identification during live events.
- Built and integrated a Jersey Number Recognition System leveraging K-Means, Deep Learning (HOG + ELM), and EasyOCR, automating the processing of 10,000+ race images with minimal manual intervention.
- Optimized backend databases and API endpoints to support high-traffic race events, reducing image retrieval and race timing processing times by 30%+.
- Partnered with the product team to seamlessly embed AI capabilities into the NOVARACE platform, directly improving the end-user experience for event organizers and participants.

RG CAR CRAZE (Coimbatore, India) November 2023 – October 2024

Role: Retail Assistant

- Interacted with customers, solved their concerns, and provided timely assistance to ensure satisfaction.
- Managed multiple tasks simultaneously, including inventory management and troubleshooting minor technical issues.
- Developed strong problem-solving abilities through handling diverse customer interactions and technical challenges.
- Performed routine system updates, backups, and maintenance tasks to ensure optimal performance and data integrity across various IT systems and applications.

PROJECTS

Coimbatore Institute of Technology (Coimbatore, India) January 2024 – April 2024

Multilingual E-commerce Voice Assistant - Project

- Built a powerful Natural Language Understanding (NLU) engine and FAQ classifier by engineering Artificial Neural Networks (ANNs) with TensorFlow/Keras and integrating Scikit-learn (TF-IDF, Naive Bayes) to achieve high-accuracy user intent recognition.
- Created a seamless, multilingual speech interface utilizing SpeechRecognition, Google Text-to-Speech (gTTS), and the Googletrans API to enable fluid, real-time voice conversations and localized product recommendations.
- Engineered a dynamic recommendation system leveraging Pandas and NumPy to parse and rank e-commerce product datasets (e.g., price, RAM, camera) based on parsed voice commands.

Coimbatore Institute of Technology (Coimbatore, India) August 2023 – December 2023

AI Fitness Tracker - Project

- Engineered a real-time pose estimation system for diverse exercises (e.g., squats, push-ups, planks) utilizing MediaPipe (BlazePose CNN) and OpenCV to track human biomechanics, ensure proper form, and automatically calculate repetitions.
- Integrated pyttsx3 to deliver real-time, dynamic text-to-speech voice feedback on user performance and exercise counts.
- Designed an automated data-logging pipeline using Python, NumPy, and SQLite to seamlessly store and track user workout metrics (counts, form, and duration) into a structured database.

Coimbatore Institute of Technology (Coimbatore, India) August 2022 – July 2023

Invest With Confidence - Project

- Engineered a predictive financial model using Keras (LSTM & Dense networks) and Scikit-learn (MinMaxScaler) to forecast next-day stock prices based on historical time-series data acquired via the YFinance API.
- Designed robust data preprocessing pipelines using Pandas and NumPy to clean and structure decades of stock market data for optimal neural network training.
- Performed real-time sentiment analysis on associated financial news by leveraging Natural Language Processing (NLP) techniques via NLTK / VADER, correlating public sentiment with market trends.

EDUCATION

Adelaide University (Adelaide, Australia) September 2025 - Present

Master of Artificial Intelligence and Machine Learning (GPA: 6.0)

Coimbatore Institute of Technology (Coimbatore, India) August 2020 - May 2024

Bachelor of Information Technology (Grade 76%)

CERTIFICATION

- Google Project Management, Coursera
- IBM Data Science Methodology, Coursera
- The Complete Python Programming Bootcamp, Udemy

LEADERSHIP & ADDITIONAL RESPONSIBILITIES

Association Treasurer (Information Technology Department), Coimbatore, India June 2023 - May 2024

Managed departmental association funds, optimizing budget allocation for academic workshops and extracurricular student events throughout the academic year.